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RESEARCH PROPOSAL ON —

“FROM SOUND TO SYNAPSES: UNRAVELLING THE COMPLEX INTERPLAY BETWEEN MUSICAL TRAINING, NEUROPLASTICITY, NEURAL ARCHITECTURE AND COGNITIVE FUNCTIONS IN THE HUMAN BRAIN”

NAME: JIGEESHA MONDAL

INSTITUTION NAME: BERKLEE COLLEGE OF MUSIC

GRADE: 1ST YEAR (FALL 2026)

COUNTRY: INDIA

RESEARCH CATEGORY: INTERDISCPLINARY STEM (MUSIC AND NEUROSCIENCE)

SUBMISSION DATE: 20.09.2026

1. Introduction and background:

1.1 Music as a complex human behaviour:

Learning to play a musical instrument is a hands-on experience that requires coordination of auditory, visual, and motor processes. Most musicians undergo long-term musical training and repeatedly practice these complex sensory and motor skills [1]. Musical training therefore provides a useful context for studying how repeated learning experiences are associated with changes in the human brain.

1.2 Musical training and the brain:

Musical training provides a useful framework for investigating how repeated learning experiences are associated with changes in the human brain. Because musical practice simultaneously engages auditory, motor, and cognitive processes, training has been associated with differences in brain structure, functional organization, and aspects of cognitive performance. These changes have been reported across different stages and durations of musical experience [2].

1.3 Neuroplasticity and experience-dependent brain changes:

- Neuroplasticity refers to the brain's ability to adapt its structure and function in response to learning new skills or to other experiences [3].
- At the cellular and network levels, experience-dependent plasticity can involve changes in synaptic strength and neural connectivity [3].
- In the human brain, neuroplasticity-related changes occur in response to different internal and external stimuli, including sensory inputs, auditory signals, cognitive exercises, and broader environmental changes [3].

1.4 Research gap and problem statement:

The difficulty is distinguishing changes connected with musical training from differences that existed before training began [2]. It is also not fully clear how structural, functional, and cognitive changes relate to one another over the course of training [3].

2. Existing Literature:

2.1 Evidence linking musical training and neuroplasticity:

Building on the concept of experience-dependent change, research demonstrates that the adult brain retains a substantial capacity to adapt in response to experience. Playing an instrument

requires the coordinated engagement of multiple sensorimotor and cognitive systems. Musical training provides a useful framework for investigating experience-dependent neural changes. The effects of musical training can be considered across three domains —

- **Structural changes**: Structural MRI studies have identified anatomical differences in gray-matter volume in motor and perisylvian regions in musicians, particularly in regions involved in auditory and motor processing [2], [3].

- **Functional changes:** Functional studies have also reported differences in the interaction between the auditory and motor networks, suggesting that musical training may influence the coordination of sensory and motor processes [2].

- **Cognitive / behavioral changes:** Musical training has additionally been associated with differences in cognitive functions, including memory, attention, and learning [2], [5].

2.2 Training effect versus pre-existing predispositions:

- **Cross-sectional evidence:** Comparisons between musicians and non-musicians have reported distinct structural and functional variations. These studies cannot isolate whether these neural differences resulted from long-term musical training or were present before training [3].

- **Longitudinal evidence:** Longitudinal studies provide stronger evidence for examining changes over time [5]. Studies of childhood musical training have reported structural changes in auditory-motor pathways along with changes in some behavioral measures [4], [5].

These findings highlight the need for longitudinal studies that examine structural, functional, and cognitive changes together.

3. Research questions:

3.1 In what ways does long-term musical training impact structural brain plasticity, specifically regarding gray-matter density and white-matter integrity?

3.2 How is sustained musical training associated with changes in functional connectivity, particularly within the auditory-motor networks?

3.3 What is the relationship between neural adaptations and measurable cognitive shifts in areas like memory and executive function?

4. Proposed Methodology:

4.1 Study design: The study will use a prospective longitudinal design to examine changes related to sustained musical training over time. Participants will be assessed at baseline and subsequent follow-up points using brain imaging and cognitive measures. This design will allow changes in brain structure, functional connectivity, and cognitive performance to be examined over the course of musical training.

4.2 Participants: The study will recruit school-aged children between 5 and 10 years of age who are beginning or participating in structured extracurricular activities. Children beginning structured instrumental music training will form the music-training group, while children participating in a structured non-musical activity will form the active-control group. Baseline characteristics will be recorded to account for pre-existing differences between the groups.

4.2.1 Musical-training group: Children enrolled in structured instrumental music training with a minimum planned exposure of approximately 2 hours per week, including formal instruction and individual practice.

The following will be recorded: • Type of instruments • frequency of lessons • practice duration • total training duration

4.2.2 Active-control group: children participating in a non-musical structured after-school activity (e.g., sports, drama, dance or any other structured activity) to control for the confounding effects of general enrichment, social engagement, and natural development.

4.3 Measurement tools:

- Structural MRI— assessment of brain morphology, including gray-matter measures.
- Diffusion Tensor Imaging (DTI)— assessment of white-matter microstructural properties.
- Functional/neurophysiological measures— functional connectivity, especially auditory-motor networks
- Cognitive testing—memory and executive function
- Training/practice questionnaires— amount and duration of musical exposure.

4.4 Data analysis:

4.4.1 Descriptive analysis: Participant characteristics, musical-training exposure, and baseline measurements will be summarised to understand the study sample.

4.4.2 Change over time: Brain and cognitive measures will be compared across the different assessment points to examine changes during the training period.

4.4.3 Group Comparisons: The music-training and active-control groups will be compared using **mixed-effects models** to examine whether their changes over time differ. **ANOVA** can be used where appropriate for simpler comparisons.

12 **4.4.4 Training Exposure:** The relationship between the amount / duration of musical training and changes in neurobiological and cognitive measures should be examined using appropriate correlation or regression analyses.

4.4.5 Cross-Domain Integration: Findings from **MRI, DTI, functional measures, and cognitive testing** will be considered together to examine whether changes in neural measures are associated with measurable changes in memory and executive function.

5. Ethical Considerations:

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- **Parental consent**: Written informed consent will be obtained from a parent or legal guardian before the child takes part in the study.
- **Child assent:** The study will be explained to the child in age-appropriate language, and their willingness to participate will be respected.

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- **Privacy and confidentiality:** Personal information and research data will be kept confidential and stored securely.
- **MRI safety:** Standard MRI screening and safety procedures will be followed before each imaging session.
- **Right to withdraw:** Participants can leave the study at any stage without penalty or disadvantage.
- **Ethics approval:** The study will begin only after approval from the relevant institutional ethics committee.

6. Expected outcomes:

The study may identify differences over time in gray-matter and white-matter organization in children receiving musical training. Changes in connectivity between brain regions involved in auditory and motor processing will be examined. Measures of memory and executive functions will be compared across the assessment periods. The analysis will explore whether the amount and duration of musical training are related to changes in neural or cognitive measures. By collecting measurements from the beginning of the study, the research may help distinguish changes occurring during the training period from differences that were already present at baseline.

7. Potential Limitations:

• Children may differ in their previous musical exposure, learning environment, family background, or other experiences before entering the study.

• The amount and consistency of practice may vary between participants, even within the music-training group.
• Some participants may not complete every assessment, which could reduce the final sample size.
• Movement during brain imaging can affect the quality of MRI data, particularly in younger children.
• Findings from children aged 5–10 years may not necessarily apply to older age groups or different populations.

8. Significance of the study: This study will examine how regular musical training may affect the developing brain. Examining the brain structure, neural connections, and cognitive abilities together may help explain how sustained musical practice is linked to memory and executive function. The findings may also provide a basis for future studies on the role of music in childhood development.

9. Project practicalities: Participant enrollment will follow strict eligibility criteria, with all baseline musical background and practice histories meticulously documented. Data acquisition will be standardized across baseline and follow-up assessments to ensure internal reliability.

10. Future research directions: Subsequent studies may expand these protocols to larger socioeconomically diverse cohorts to enhance external validity. Long-term follow-up designs are needed to evaluate the durability of neural adaptations linked with early musical training.

11. Conclusion: The study will examine the longitudinal relationship between childhood musical training, neurostructural changes and cognitive development. By employing an active-control design, the research aims to clarify how structured musical engagement shapes neurodevelopmental trajectories.

12. Broader research value: Music is more than an artistic activity; it is also a complex form of learning that engages several systems of the brain at the same time. Studying musical training can therefore provide a useful way to understand how repeated experience and skill development shape the brain. Looking at brain structure, function, and cognitive performance together may build a more complete picture of experience-dependent neuroplasticity.

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